

Welcome to **instats**

The session will begin shortly.

Start

SECOND MOVEMENT, 30 MINUTES

The Colab tour.

A free Python environment in your browser, with a GPU attached. You will use it during this workshop. You may use it for the rest of your career.

02

Colab is a notebook, plus a remote runtime, plus a hardware menu.

LAYER 01

The notebook

Cells of code and prose, like an Rmd file but interactive. Click a cell, press Shift Enter, output appears beneath it.

LAYER 02

The remote runtime

The runtime is on Google's servers. Press run, the code executes there, you see the output in your browser.

LAYER 03

The hardware menu

Pick CPU, GPU, or TPU. Free tier gives a CPU plus an occasional T4 GPU. Paid Colab Pro reliably attaches a beefier card.

For an encoder workflow, that GPU is roughly 10 to 50 times faster than CPU.

Five things to find on your screen.

- **Code cells.** Type Python, press Shift Enter.
- **Text cells.** Markdown for notes and headings.
- **Runtime menu.** Pick CPU or GPU. Restart runtime when things misbehave.
- **Files panel on the left.** Upload data. Mount Google Drive.
- **Secrets, the key icon.** Where API keys live. Never paste a key into a code cell.

THE RIGHT WAY TO READ A SECRET

```
from google.colab import userdata
import os

os.environ["ANTHROPIC_API_KEY"] = (
    userdata.get("ANTHROPIC_API_KEY")
)
```

Add the key via the left-sidebar key icon. Name it ANTHROPIC_API_KEY. We use this pattern on Day 3 with the Anthropic API.

Three lines of Python load a state-of-the-art encoder.

```
# cell 1
!pip install -q transformers

# cell 2
from transformers import pipeline
clf = pipeline(
    "sentiment-analysis",
    model="cardiffnlp/twitter-roberta"
        "-base-sentiment-latest",
)

# cell 3
clf("Today's bipartisan vote was a real step forward.")
# [{'label': 'positive', 'score': 0.94}]
```

WHAT THE AUDIENCE SEES LIVE

- One install, about 30 seconds.
- One pipeline construction, about 20 seconds for first model download.
- Five inference calls on five sentences, about a second each.

A 125-million-parameter Transformer running on Google's hardware in twenty seconds. Five years ago this required a graduate course. Today it is a pipeline call.

The number on the right is what we paid for.



These numbers come from the live notebook in the last two minutes. We do not quote pre-baked numbers. Once you understand the speedup, you stop avoiding tasks that take hours. You run them while you make coffee.

Numbers above are illustrative ratios. Live values fill the slide once the demo cell finishes.

Questions about Colab itself.

COMMON QUESTIONS, PRE-ANSWERED

- **Do I need a paid account?** No. Free tier handles every notebook in this workshop. Day 2 batch demo is slower without GPU.
- **Can I run on my laptop?** Yes if you are comfortable installing Python and CUDA. For today, just use Colab.
- **What about Drive storage?** Free Drive is 15 GB. Plenty for our notebooks. Datasets bigger than 5 GB warrant the Pro plan.

THINGS WE WILL DEFER

- Fine-tuning. Day 2 mentions briefly. Beyond scope of this workshop.
- Local installs. Out of scope. Use Colab.
- Project-specific replication paths. Office hours.

THIRD MOVEMENT, 35 MINUTES

Stance on real UN speeches.

No fine-tuning yet. Off-the-shelf model, real text, real measurement question.

03

The UN General Debate Corpus.

Eight thousand speeches, 1970 to present.

SPECS

- **Source.** Harvard Dataverse, doi:10.7910/DVN/OTJX8Y.
- **License.** CC0. Fully open.
- **Coverage.** Heads of delegation, opening of the UN General Assembly.
- **For today.** The 2017-only snapshot mirrored in `quanteda.corpora`. About 196 speeches, 1 MB.
- **Sample.** Sentence-tokenize, take roughly 200 sentences across countries.

THREE SAMPLE SENTENCES

"My country reaffirms its commitment to the Paris Agreement and calls on all states to raise their nationally determined contributions."

"Investment in fossil-fuel infrastructure remains a national priority for energy security and industrial development."

"The reform of the Security Council remains an urgent task that this Assembly must not defer for another decade."

The encoder will read each, score it, and give us back a stance label. Public corpus. Permissive license. Substantive question political scientists actually ask.

One model. Three plain-English labels.

THE MODEL

MoritzLaurer/DeBERTa-v3-base-mnli-fever-anli

- **License.** MIT. Open weights.
- **Size.** ~184 million parameters, ~735 MB.
- **Training.** Three NLI corpora, millions of premise-hypothesis pairs.
- **Pipeline.** zero-shot-classification.

CANDIDATE LABELS

```
candidate_labels = [  
    "the speaker supports stronger  
    international climate action",  
    "the speaker is skeptical of  
    international climate action",  
    "the speaker does not  
    discuss climate",  
]
```

We are not training. We are giving the model three plain-English statements and asking which one each sentence entails. The argmax of the three returned probabilities is the prediction.

Three things to watch for.

WATCH 01

Easy hits

Sentences that mention Paris Agreement, carbon emissions, or fossil fuels resolve cleanly to the supportive label, often above 0.85.

Sentences about peacekeeping or Security Council reform land in the off-topic bucket.

WATCH 02

Hidden climate language

Sentences about "energy security" or "industrial transition" can carry a real climate stance the model misreads as off-topic.

Sentences about "rising seas" land correctly but at low confidence. The frame is implicit, not lexical.

WATCH 03

Calibration

The model returns probabilities. Even confident-looking labels can be 0.55 versus 0.45 splits.

We will plot the probability histogram and discuss what cutoff a real paper would use.

Live demo, about 18 minutes in the notebook. Drop confusions in chat as the cells run.

Where off-the-shelf zero-shot stumbles.

PATTERN	WHAT YOU SEE	DIAGNOSIS
Hidden lexicon	"Energy independence" without a climate keyword lands off -topic.	The model has no climate -relevant lexicon attached.
Mixed paragraph	Praise for Paris implementation that opens with criticism of slow progress lands skeptical.	Model picks the first cue and misses the second.
Low -confidence argmax	Aspirational language at 0.51 in the supportive bucket.	Argmax is what we read. Look at probabilities, not labels.
Implicit framing	"Rising seas" lands supportive but at low confidence.	Climate is implicit, not lexical.
Rhetoric	Sarcasm or hedging in formal UN register lands the wrong way.	Encoders are weak on tone -of -voice.

Day 2 is the answer to all five. Hand-coded gold set. Validation. A confusion matrix you can put in an appendix.

FOURTH MOVEMENT, 15 MINUTES

Your turn.

Open free Colab. Run a tiny version yourself. The goal is to know that you can.

04

Open the notebook. Run all. Then play.

SETUP

- Open the hands-on link from chat. Sign into Google if asked.
- **File, Save a copy in Drive.** Keep your edits.
- **Runtime, Run all.** First cell installs transformers, ~30 sec. Second cell downloads model, ~45 sec. Then seconds per cell.
- Total runtime under three minutes on free CPU.

THREE EXPERIMENTS

- **Experiment 01.** Read printed outputs for ten UN speech sentences. Notice which are easy and which are mixed.
- **Experiment 02.** Edit the candidate labels. Try shorter labels. Add a fourth label. See how output depends on wording.
- **Experiment 03.** Replace one example with a sentence from your own research. The model has never seen your data. Often it still produces a useful first cut.

Drop confusions in chat. We surface a few during wrap-up.

Four common issues.

SYMPTOM	WHAT PROBABLY HAPPENED	FIX
"RESTART REQUIRED" warning after pip install.	The runtime needs a restart to pick up the new package.	Runtime, Restart, then run all from the top.
Model download stalls.	Hugging Face mirror is slow.	Wait sixty seconds and re-run the cell.
"CUDA out of memory" on GPU.	An earlier cell loaded a too-large model.	Runtime, Disconnect and delete runtime. Start fresh.
Cells run, output looks wrong.	You edited labels but did not re-run upstream cells.	Runtime, Run all. Re-runs in dependency order.

If none of these match, paste the last 6 lines of your error into chat.

What you have actually done in the last two and a half hours.

- You can name the two model families and articulate why a researcher chooses one over the other.
- You can read the NLI premise-hypothesis trick as a universal classifier and design candidate labels for your own task.
- You have run a state-of-the-art encoder on Google Colab, on free hardware, against real political text.
- You have seen what off-the-shelf can and cannot do. Day 2 closes the gap with validation against a hand-coded gold set.

If a colleague asks you tomorrow whether they should use ChatGPT to label their tweets, you now have a real answer.

TOMORROW, DAY 02

From a model to a measurement.

We take the same encoder, run it across the full UN General Debate Corpus, build a hand-coded gold set, and turn predictions into a defensible proxy.

HOMEWORK

Re-run the hands-on notebook on your own. Try one labeling task from your own research, even if rough. Bring three examples to Day 2.

OPTIONAL READING

Bestvater & Monroe (2023, PA), "Sentiment Is Not Stance." The single best 30-page primer for the validity questions Day 2 asks.

Stop